Chapter 25 Review

Campbell Biology, AP Edition, 12e

Chapter 25 Review

Summary of Key Concepts

How does one evaluate evidence from fossils, morphological differences, and/or biochemical differences to determine an evolutionary connection between organisms that span time and space? **(BIG IDEA 1)**

Concept 25.1: Conditions on early Earth made the origin of life possible

- Experiments simulating possible early atmospheres have produced organic molecules from inorganic precursors. Amino acids, lipids, sugars, and nitrogenous bases have also been found in meteorites.
- Amino acids and RNA nucleotides polymerize when dripped onto hot sand, clay, or rock. Organic compounds can spontaneously assemble into **protocells**, membrane-bounded droplets that have some properties of cells.
- The first genetic material may have self-replicating, catalytic RNA. Early protocells containing such RNA would have increased through natural selection.

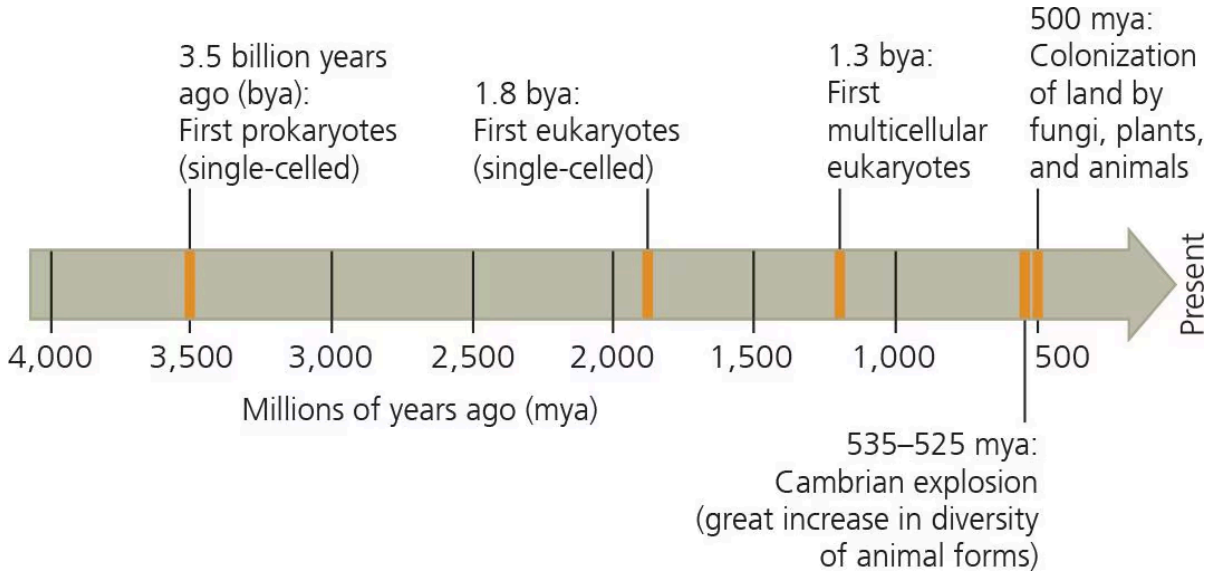
Describe the roles that montmorillonite clay and vesicles may have played in the origin of life.

Concept 25.2: The fossil record documents the history of life

- The fossil record, based largely on fossils found in sedimentary rocks, documents the rise and fall of different groups of organisms over time.
- Sedimentary strata reveal the relative ages of fossils. The ages of fossils can be estimated by **radiometric dating** and other methods.
- The fossil record shows how new groups of organisms can arise via the gradual modification of preexisting organisms.

What are the challenges of estimating the ages of old fossils? Explain how these challenges may be overcome in some circumstances.

Concept 25.3: Key events in life's history include the origins of unicellular and multicellular organisms and the colonization of land



Events are as follows:

- 3500 million years ago (3.5 billion years ago): first prokaryotes (single-celled).
- 1800 million years ago (1.8 billion years ago): First eukaryotes (single-celled).
- 1200 million years ago (1.2 billion years ago): First multicellular eukaryotes.
- 535 to 525 million years ago: Cambrian explosion (great increase in diversity of animal forms).
- 500 million years ago: Colonization of land by fungi, plants and animals.

What is the "Cambrian explosion," and why is it significant?

Concept 25.4: The rise and fall of groups of organisms reflect differences in speciation and extinction rates

- In **plate tectonics,** continental plates move gradually over time, altering the physical geography and climate of Earth, leading to extinctions in some groups and speciation in others.
- Evolutionary history has been punctuated by five ****__mass extinctions__ that radically altered life's history. Possible causes for these extinctions include continental drift, volcanic activity, and impacts from asteroids.
- Large increases in the diversity of life have resulted from ****__adaptive radiations__ that followed mass extinctions. Adaptive radiations have also occurred in groups of organisms that possessed major evolutionary innovations or that colonized new regions in which there was little competition from other organisms.

Explain how the broad evolutionary changes seen in the fossil record are the cumulative result of speciation and extinction events.

Concept 25.5: Major changes in body form can result from changes in the sequences and regulation of developmental genes

- Developmental genes affect morphological differences between species by influencing the rate, timing, and spatial patterns of change in an organism's form as it develops into an adult.
- The evolution of new forms can be caused by changes in the nucleotide sequences or regulation of developmental genes.

How could changes in a single gene or DNA region ultimately lead to the origin of a new group of organisms?

Concept 25.6: Evolution is not goal oriented

- Novel and complex biological structures can evolve through a series of incremental modifications, each of which benefits the organism that possesses it.
- Evolutionary trends can be caused by natural selection in a changing environment or species selection, resulting from interactions between organisms and their current environments.

Explain the reasoning behind the statement "Evolution is not goal oriented."